assignment of utilities should obey; for example, if an agent prefers *A* to *B* and also prefers *B* to *C*, they should definitely prefer *A* to *C*. Anyone who goes through life violating the axioms of decision theory is deemed to be irrational, and that's that.

To use this framework in the context of Many-Worlds, we ask how a rational agent should behave, knowing that the wave function of the universe was about to branch and knowing what the amplitudes of the different branches were going to be. For example, an electron in an equal superposition of spin-up and spin-down is going to travel through a Stern-Gerlach magnet and have its spin be measured. Someone offers to pay you $2 if the result is spin-up, but only if you promise to pay them $1 if the result is spin-down. Should you take the offer? If we trust the Born rule, the answer is obviously yes, since our expected payoff is 0.5($2) + 0.5(-$1) = $0.50. But we're trying to derive the Born rule here; how are you supposed to find an answer knowing that one of your future selves will be $2 richer but another one will be $1 poorer? (Let's assume you're sufficiently well-off that gaining or losing a dollar is something you care about, but not life-changing.)

The manipulations are trickier here than in the previous case where we were explaining probabilities as credences in a situation of self-locating uncertainty, so we won't go through them explicitly, but the basic idea is the same. First we consider a case where the amplitudes on two different branches are equal, and we show that it's rational to calculate your expected value as the simple average of the two different utilities. Then suppose we have an unbalanced state like Ψ above, and I ask you to give me $1 if the spin is measured to be up and promise to give you $1 if the spin is down. By a bit of mathematical prestidigitation, we can show that your expected utility in this situation is exactly the same as if there were three possible outcomes with equal amplitudes, such that you give me $1 for one outcome and I give you $1 for the other two. In that case, the expected value is the average of the three different outcomes.

At the end of the day, a rational agent in an Everettian universe acts precisely *as if* they live in a nondeterministic universe where probabilities are given by the Born rule. Acting otherwise would be irrational, if we accept the various plausible-seeming axioms about what it means to be rational in this context.